

IIT-JEE 2005 TEST PAPER

MAIN EXAMINATION SUBJECT : PHYSICS

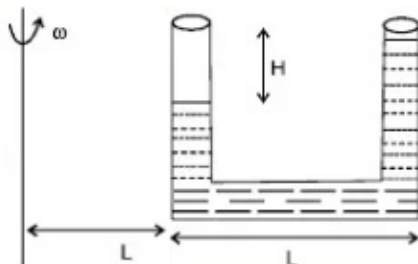
Time : 2.00 Hrs

Max. Marks : 60

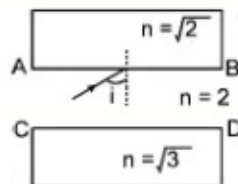
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- A train is passing a stationary observer at station with constant velocity. If the frequency observed by the person during its approach and recession are 2.2 kHz and 1.8 kHz respectively. Then find the velocity of train if the velocity of sound in air is 300 m/s.
- Side of a cube is measured with the help of vernier calliper. Main scale reading is 10 mm and vernier scale reading is 1. It is known that 9 M.S.D. = 10 V.S.D.. Mass of the cube is 2.735 g. Find density of the cube upto appropriate significant figure.
- Find height difference H of the liquid column in two limbs when the U tube is rotated with angular speed ω . (while diameter of the tube $d \ll L$)



- What is the minimum angle of incidence of incident ray at the surface AB so that the light is totally reflected at the surfaces AB and CD.



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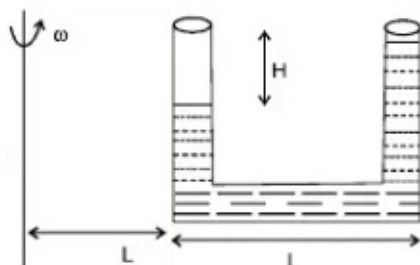
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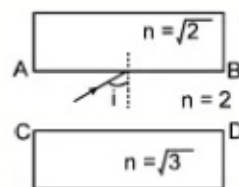
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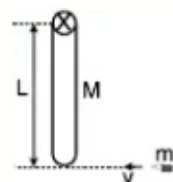
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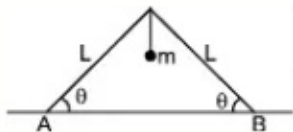
4. What is the minimum angle of incidence of incident ray at the surface AB so that the light is totally reflected at the surfaces AB and CD.



5. A rod of mass M , length L hinged at its one end is in vertical equilibrium position. A bullet of mass m , moving with velocity v strikes the lower end of the rod and gets embedded into it. Find the angular velocity of the rod just after the collision.



6. A bubble of conducting liquid is charged to potential v , it has radius a and thickness $t \ll a$. It collapses to form a droplet. Find potential of the droplet.
 7. A solid cylinder of mass ' m ' rolls without slipping on a fixed long inclined plane inclined at an angle θ with horizontal. Find the linear acceleration of the centre of mass of the cylinder.
 8. A transverse wave travelling in a string produces maximum transverse velocity of 3 m/s and maximum transverse acceleration 90 m/s^2 in a particle. If the velocity of wave in the string is 20 m/s . Determine the wave form?
 9. Two rods of mass M and length L are placed as shown in figure. If the system is in equilibrium then find the magnitude and direction of frictional force at the points of contact A and B ?
- 
- The diagram shows two rods of length L and mass M each, inclined at an angle θ to the horizontal. They meet at a top vertex where a mass m is suspended. The left rod touches the ground at point A, and the right rod touches the ground at point B. The ground is represented by a horizontal line.



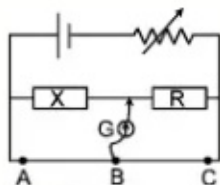
10. The potential energy of a mass 'm' is given by the following relation

$$U = E_0 \text{ for } 0 \leq x \leq 1 = 0 \text{ for } x > 1$$

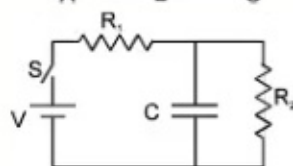
If λ_1 and λ_2 are the de-Broglie wavelengths of the mass in the region $0 \leq x \leq 1$ and for $x > 1$ respectively and the

total energy be $2E_0$, then find the value of $\frac{\lambda_1}{\lambda_2}$?

11. For the three values of resistances R namely R_1 , R_2 and R_3 the balanced positions of jockey are at A, B and C respectively. Which position will show most accurate result for calculation of X . Give reason. B is near the mid point of the wire.



12. In the given circuit the capacitor C is uncharged initially and switch 'S' is closed at $t = 0$. If charge on capacitor at time 't' is given by equation $Q = Q_0 (1 - e^{-\alpha t})$. Find value of Q_0 and α ?

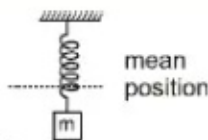


13. A metal target consist of large number of atoms (with each atom having number of neutrons is 30). The radius ratio of the target atom to ${}^4_2\text{He}$ is $(14)^{1/3}$.

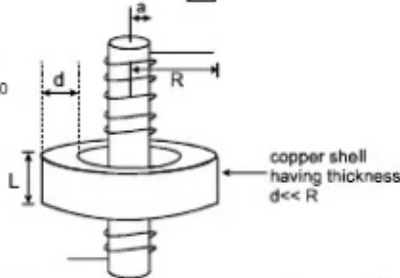
- Find the atomic number of metal
- Find the frequency of K_{α} X-ray for that metal

(Given : Rydberg constant = $1.1 \times 10^7 \text{ m}^{-1}$; speed of light = $3 \times 10^8 \text{ m/s}$.)

14. A block is performing SHM of amplitude 'A' in vertical direction. When block is at ' y_0 ' (measured from mean position), it detaches from spring, so that spring contracts and does not affect the motion of the block. Find ' y_0 ' such that block attains maximum height from the mean position. (Given $A\omega^2 > g$)



15. Current passing through a long solenoid having n turns per unit length is $I = I_0 \sin \omega t$. Find induced current through copper shell having resistivity ρ as shown in figure.



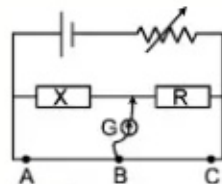
16. A mass of 1 kg at 20°C is given an energy 20000 J at 1 atmospheric pressure

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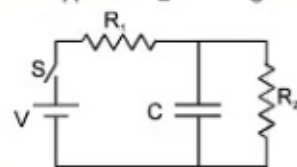
$$U = E_0 \text{ for } 0 \leq x \leq 1 = 0 \text{ for } x > 1$$

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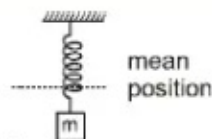


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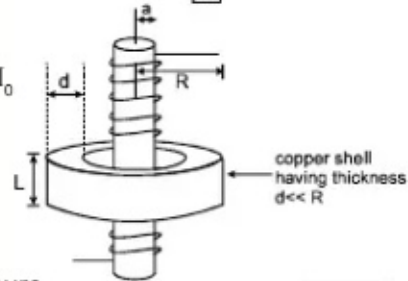
- (a) Find the atomic number of metal
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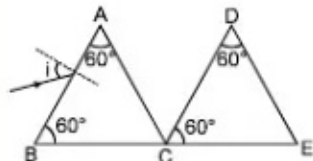
16. A mass of 1 kg at 20°C is given an energy 20000 J at 1 atmospheric pressure

- (i) Find the change in temperature of mass. (ii) Find the work done by mass.
(iii) Find the increase in internal energy of mass.

{Given $1 \text{ atm} = 10^5 \text{ N/m}^2$, Specific heat capacity = $400 \text{ J/kg } ^\circ\text{C}$, density = 9000 kg/m^3 , coefficient of cubical expansion = $9 \times 10^{-5} / ^\circ\text{C}$ }

17. In the figure two triangular prisms are shown each of refractive index $\sqrt{3}$.

- (a) Find the angle of incidence on the face AB for minimum deviation from the prism ABC?
- (b) Find the angle through which the prism DCE should be rotated about the edge passing through point C so that there should be minimum deviation from the system?



18. Relation for a Galvanometer having number of turns N , area of cross section A and moment of inertia I is given as : $\tau = Ki$ where K is a positive constant and ' i ' is current in the coil placed in the magnetic field B .
- (i) Find K in terms of B , N and A
- (ii) Find torsional constant of spring if a current I_0 produces a deflection of $\frac{\pi}{2}$
- (iii) If an instant charge Q is flown through the galvanometer, find the maximum deflection in the coil.

ANSWER KEY TO IIT-JEE 2005 TEST PAPER

MAIN EXAMINATION

- | | | | | | | | |
|----|------------------------|----|---------|----|----------------------------------|----|----------------|
| 1. | $V_s = 30 \text{ m/s}$ | 2. | 0.00265 | 3. | $H = \frac{\omega^2 \ell^2}{2g}$ | 4. | $i > 60^\circ$ |
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5. $\omega = \frac{3mv}{(M+3m) \times L}$ 6. $v' = v \left(\frac{a}{3t} \right)^{1/3}$ 7. $a = \frac{2g \sin \theta}{3}$

8. Equation of wave in string $y = 0.1 \sin \left(30t \pm \frac{3}{2}x + \phi \right)$ [where ϕ is initial phase]

9. $f = \frac{(M+m)g \cot \theta}{2}$ required friction 10. $\sqrt{\frac{2E_0}{E_0}} = \sqrt{2}$

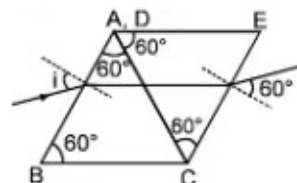
11. Fractional error in x is least if $(100 - \ell) \ell$ is maximum and it is when $\ell = 50$ cm.

12. $Q_0 = \frac{R_2 VC}{R_1 + R_2}$ & $\alpha = \frac{(R_1 + R_2)}{CR_1 R_2}$ 13. (a) 26 (b) $154875 \times 10^{12} \text{ Hz}$

14. $y_0 = \frac{g}{\omega^2}$ 15. $\frac{(\mu_0 n a^2 I_0 \omega \cos \omega t) L d}{2 \rho R}$

16. (i) 50°C (ii) 0.05 J (iii) 19999.95 J

17. (a) Angle of incidence = $i = 60^\circ$
 (b) For minimum deviation for the system, deviation from two prism will be zero and the two prism will form parallel slab for this situation prism DCE is rotated in anticlockwise direction by 60°



18. $\theta = Q \sqrt{\frac{TNAB}{2I\mu_0}}$

Useful data

Useful Data :

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Gas constant, $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1} = 8.314 \text{ J}^{-1} \text{ K}^{-1} \text{ mol}^{-1}$

1 Farady constant $= 96.500 \text{ C mol}^{-1}$

Avagadro number, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

Planck constant, $h = 6.626 \times 10^{-34} \text{ Js}$

Speed of light, $c = 3.00 \times 10^8 \text{ ms}^{-1}$

Rydberg constant for hydrogen, $R_H = 109679 \text{ cm}^{-1}$

Atomic number : $N = 7$; $O = 8$; $P = 15$; $Co = 27$; $Ni = 28$; $Xe = 54$.

1. If after complete ozonolysis of one mole of monomer of natural polymer gives two moles of CH_2O and one mole

$\begin{array}{c} \text{CH}_3 \\ | \\ \text{O} = \text{C} - \text{CH} = \text{O} \end{array}$

of $\text{O} = \text{C} - \text{CH} = \text{O}$. Identify the monomer and draw the all-cis structure of natural polymer.

2. Complete the following equation :



3. (a) What amount of CaO in grams is required to neutralise 852 g of P_4O_{10} ?

(b) Write the structure of P_4O_{10} .

4. In a FCC lattice of a metal, edge length is 400 pm. Find the maximum diameter of an atom which can be accommodated in an interstitial gap in this lattice without causing any distortion.

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Atomic number : N = 7; O = 8; P = 15; Co = 27; Ni = 28; Xe = 54.

- If after complete ozonolysis of one mole of monomer of natural polymer gives two moles of CH_2O and one mole

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2. Complete the following equation :

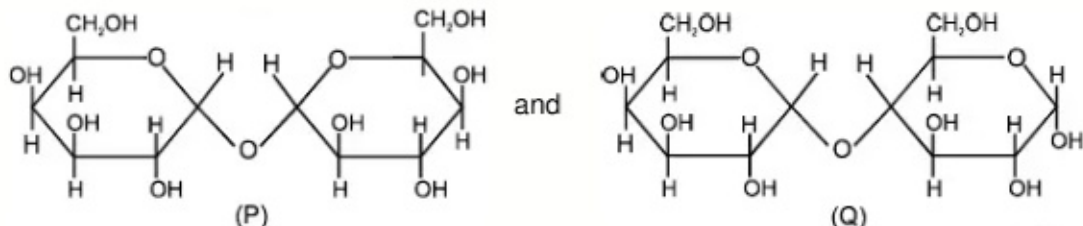


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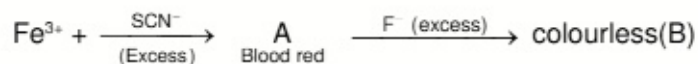
Which of (P) and (Q) will not give positive Tollen's test ? Explain.

9.

Give equations and describe the process for developing of black and white photographic film. When sodium thiosulphate solution is treated with acidic solution turns milky white. Give the half reactions of the above described process.

10.

In the given reaction sequence, Identify (A) and (B)



- (a) Write the IUPAC name of (A) and (B).
 (b) Find out the spin only magnetic moment of B.

11. For a reaction $2\text{X(g)} \longrightarrow 3\text{Y(g)} + 2\text{Z(g)}$ the following data is obtained.

Time (min)	P _x (mm of Hg) (Partial pressure of X)
0	800
100	400
200	200

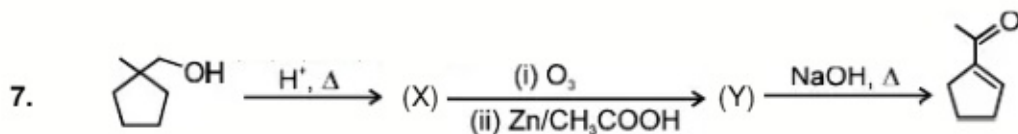
Find order with respect to X, rate constant, time taken for 75% completion and find the total pressure when partial pressure of X, P_x = 700 mm of Hg.

12.

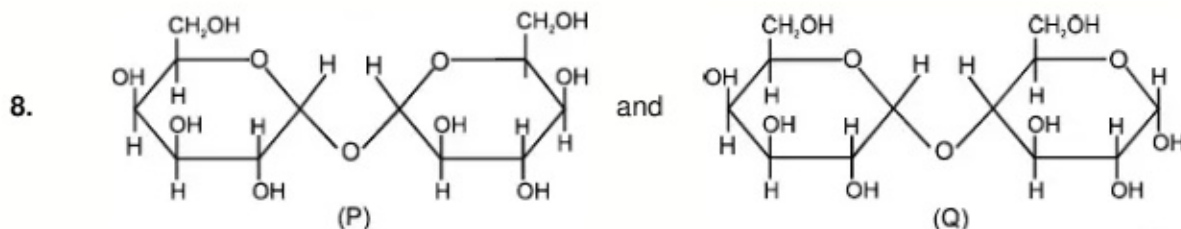
(a) Using Bohr's model for hydrogen atom, find the speed of electron in the first orbit if the Bohr's radius is $a_0 = 0.529 \times 10^{-10} \text{ m}$. Find de-Broglie wavelength of the electron also.

(b) Find the orbital angular momentum of electron if it is in 2p orbital of H in terms of $\frac{h}{2\pi}$.

5. 20% of surface sites are occupied by N_2 molecules. Number of surface sites per unit area is $6.023 \times 10^{14} \text{ cm}^{-2}$ and total area of catalyst surface is 1000 cm^2 . Now when catalyst is heated to 300K N_2 gas desorbed and evolved gas occupied, 2.46 cm^3 at 0.001 atm . Find the no. of sites occupied by each molecule of N_2 .
6. Predict whether the following molecules are iso-structural or not. Justify your answer.
 (i) NMe_3 (ii) $\text{N}(\text{SiMe}_3)_3$

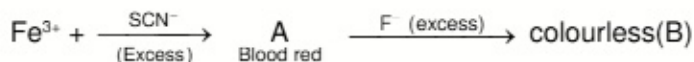


Identify X and Y.



Which of (P) and (Q) will not give positive Tollen's test? Explain.

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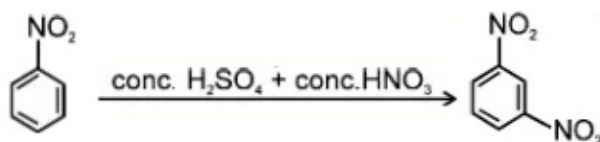
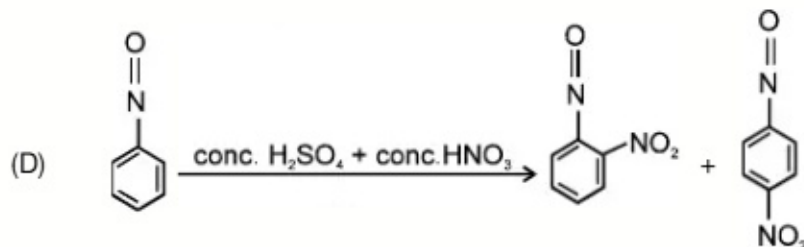
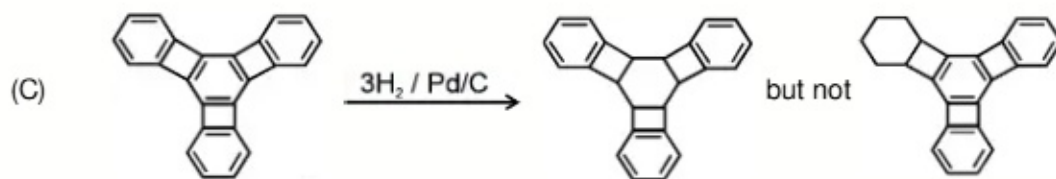
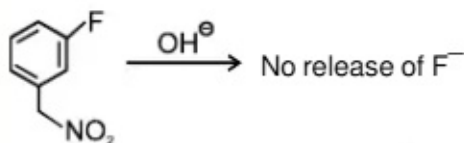
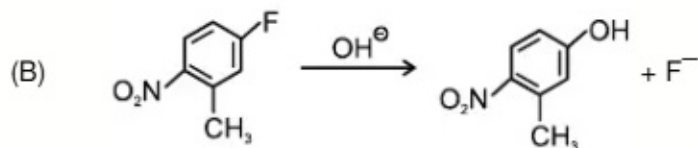
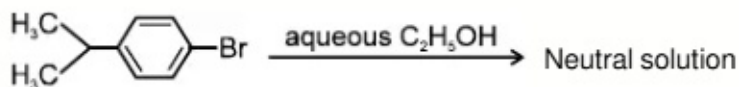


- (a) Write the IUPAC name of (A) and (B).
 (b) Find out the spin only magnetic moment of B.
11. For a reaction $2\text{X(g)} \longrightarrow 3\text{Y(g)} + 2\text{Z(g)}$ the following data is obtained.

13. $X(C_5H_{13}N) \xrightarrow[(-N_2)]{NaNO_2 + HCl} (Y) + \text{other products.}$
 (optically active) tertiary alcohol

Identify X and Y. State whether Y is optically active or not. Also draw the structures of all the intermediates (if any).

14. Explain the following observations



17. It is given that for conformational isomers, the net dipole moment is

$$\mu_{\text{obs}} = \sum \mu_i x_i$$

where μ_{obs} = observed dipole moment of the compound

μ_i = dipole moment of the stable conformational isomers

x_i = mole fraction of stable conformers

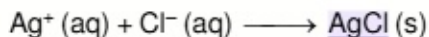
for the compound $Z - \text{CH}_2 - \text{CH}_2 - Z$ draw the Newman projection formula of all the stable conformational isomers, if $\mu_{\text{obs}} = 1\text{D}$, and $x_{\text{anti}} = 0.82$, then find μ_{gauche} .

Now draw the Newman projection formula of the most stable conformation of meso $Y - \text{CHD} - \text{CHD} - Y$

(a) If Y is CH_3 (rotation about $\text{C}_2 - \text{C}_3$ bond).

(b) If Y is OH (rotation about $\text{C}_1 - \text{C}_2$ bond).

18. (a) Calculate ΔG°_f of the following reaction :



Given : $\Delta G^\circ_f (\text{AgCl}) = -109 \text{ kJ/mole}$, $\Delta G^\circ_f (\text{Cl}^-) = -129 \text{ kJ/mole}$, $\Delta G^\circ_f (\text{Ag}^+) = 77 \text{ kJ/mole}$

Represent the above reaction in form of a cell

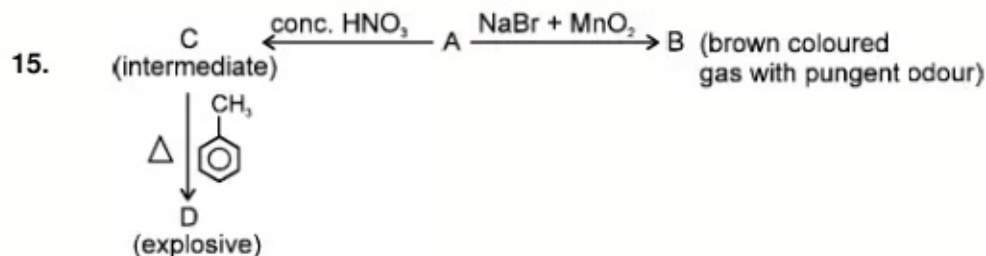
Calculate E° of the cell. Also, find $\log_{10} K_{\text{SP}}$ of AgCl .

(b) $6.539 \times 10^{-2} \text{ g}$ of metallic Zn ($\text{amu} = 65.39$) was added to 100 ml of saturated solution of AgCl .

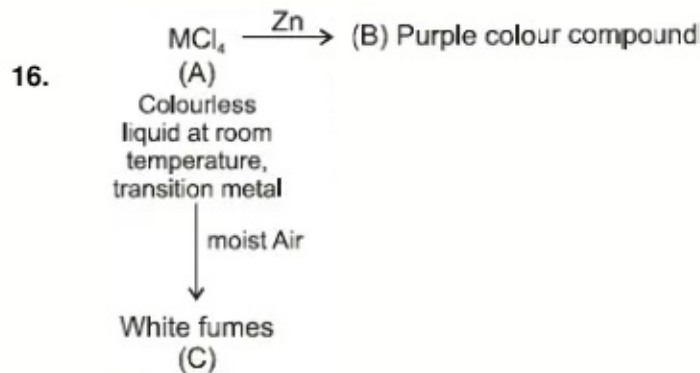
Calculate $\log_{10} \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$, at equilibrium given that



Also, find how many moles of Ag will be formed ?



Identify A, B, C, D and write the balanced chemical equation of formation of A to B and A to C.



Identify (A), (B) and (C). Also explain colour difference between (A) and (B).

17. It is given that for conformational isomers, the net dipole moment is

$$\mu_{\text{obs}} = \sum \mu_i x_i$$

where μ_{obs} = observed dipole moment of the compound

μ_i = dipole moment of the stable conformational isomers

x_i = mole fraction of stable conformers

for the compound $\text{Z} - \text{CH}_2 - \text{CH}_2 - \text{Z}$ draw the Newman projection formula of all the stable conformational isomers, if $\mu_{\text{obs}} = 1\text{D}$, and $x_{\text{anti}} = 0.82$, then find μ_{gauche} .

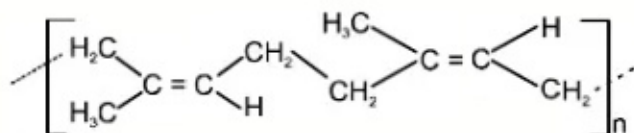
Now draw the Newman projection formula of the most stable conformation of meso $\text{Y} - \text{CHD} - \text{CHD} - \text{Y}$

(a) If Y is CH_3 (rotation about $\text{C}_2 - \text{C}_3$ bond).

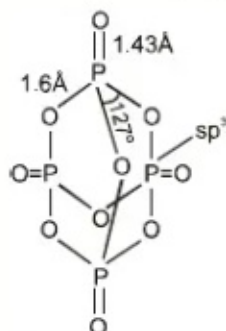
(b) If Y is OH (rotation about $\text{C}_1 - \text{C}_2$ bond).

MAIN EXAMINATION

1. Monomer of the polymer is $\text{CH}_2 = \overset{\text{CH}_3}{\underset{|}{\text{C}}} - \text{CH} = \text{CH}_2$
All cis polymer is



2. (a) $2 \frac{1}{2}n$, (b) $^{84}_{36}\text{Kr}$
3. (a) 1008 g.
(b) Structure of P_4O_{10} .

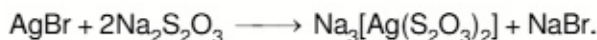


4. 117.08 pm.
5. 2
6. Not isostructural.

7. and $\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - (\text{CH}_2)_4 - \text{CH} = \text{O}$
(X) (Y)

8. P will not give positive Tollen's test.

9.
Hydroquinone Quinone



10. (a) (A) = Pentaquaithiocyanato-S-iron(III) ; (B) = Hexafluoridoferrate(III) (b) 5.93 B.M.

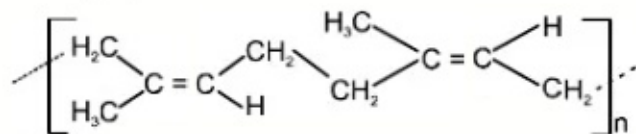
11. Order = 1, Rate constant = $6.93 \times 10^{-3} \text{ min}^{-1}$, time for 75% completion = 200 min.
Total pressure = 950 mm Hg.

ANSWER KEY TO IIT-JEE 2005 TEST PAPER

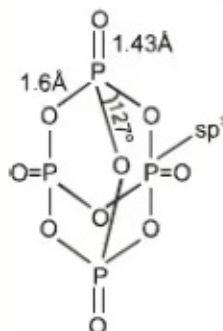
MAIN EXAMINATION

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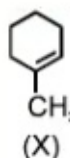
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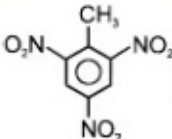
8. P will not give positive Tollen's test.

12. (a) $3.32 \times 10^{-10} \text{ m}$ (b) $\sqrt{2} \cdot \left(\frac{h}{2\pi} \right)$

Speed = $2.186 \times 10^6 \text{ m/sec}$.

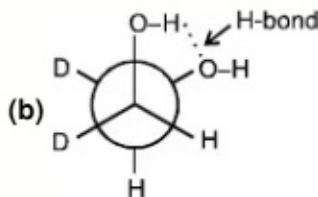
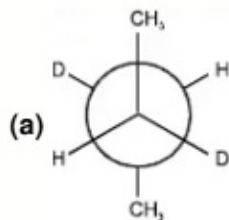
13. $\text{CH}_3 - \text{CH}_2 - \underset{\text{H}}{\overset{\text{CH}_3}{\text{C}^*}} - \text{CH}_2 - \text{NH}_2$ and $\text{CH}_3\text{CH}_2 - \underset{\text{OH}}{\overset{\text{CH}_3}{\text{C}}} - \text{CH}_3$
 (X) (Y)
 (Optically inactive)

14. (A) In 1st SN^1 reaction is possible so by-product is HBr in 2nd SN^1 reaction is not possible.
 (B) 1st can give SN^2 Ar but 2nd can not give because -m of $-\text{NO}_2$ is not operating.
 (C) 2nd product has two antiaromatic rings but 1st does not have antiaromatic system.
 (D) $-\text{NO}_2$ is metadirecting but $-\text{N}=\text{O}$ group is ortho-para directing due to +m of $-\text{N}=\text{O}$.

15. (A) H_2SO_4 (conc.), (B) Br_2 , (C) $\text{NO}_2^\oplus$, (D)  (T.N.T. = Trinitrotoluene)

16. (A) = TiCl_4 ; (B) = $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_3$; (C) = HCl
 (B) Ti^{3+} has one unpaired electron so coloured while (A) Ti^{4+} is diamagnetic so colourless.

17. $\mu_{\text{(Gauche)}} = 5.55 \text{ D}$



18. (a) $E^\circ = 0.59 \text{ V}$, $\log_{10} K_{\text{SP}} = -10 \Rightarrow K_{\text{SP}} = 10^{-10}$.
 cell representation: $\text{Ag}(\text{s}) \mid \text{AgCl}(\text{s}) \mid \text{Cl}^-(\text{aq}) \parallel \text{Ag}^+(\text{aq}) \mid \text{Ag}$.
 (b) $52.8, 10^{-6}$ moles of Ag is precipitated.

IIT-JEE 2005 TEST PAPER

MAIN EXAMINATION SUBJECT : MATHEMATICS

Time : 2.00 Hrs

Max. Marks : 60

GENERAL INSTRUCTIONS

1. There are 18 questions in this paper. Attempt ALL the questions.
 2. Question Number 1 to 8 carry 2 Marks each and Question Number 9 to 16 carry 4 Marks each and Question Number 17 to 18 carry 6 Marks each.
 3. The use of Arabic numerals (0,1,2,.....9) only is allowed in answering the questions irrespective of the language in which you answer.
-

1. Circles with radii 3, 4 and 5 touch each other externally. If P is the point of intersection of tangents to these circles at their points of contact, find the distance of P from the points of contact.
2. Evaluate $\int_0^{\pi} e^{|\cos x|} \left(2 \sin \left(\frac{1}{2} \cos x \right) + 3 \cos \left(\frac{1}{2} \cos x \right) \right) \sin x \, dx$.
3. If total number of runs scored in n matches is $\left(\frac{n+1}{4} \right) (2^{n+1} - n - 2)$ where $n > 1$, and the runs scored in the k^{th} match are given by $k \cdot 2^{n+1-k}$, where $1 \leq k \leq n$, find n
4. A person goes to office either by car, scooter, bus or train, the probability of which being $\frac{1}{7}, \frac{3}{7}, \frac{2}{7}$ and $\frac{1}{7}$ respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is $\frac{2}{9}, \frac{1}{9}, \frac{4}{9}$ and $\frac{1}{9}$ respectively. Given that he reached office in time, then what is the probability that he travelled by a car.
5. Find the range of values of 't' for which $2 \sin t = \frac{1-2x+5x^2}{3x^2-2x-1}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$.
6. The area of the triangle formed by the intersection of a line parallel to x-axis and passing through P(h, k) with the lines $y = x$ and $x + y = 2$ is $4h^2$. Find the locus of the point P.
7. If $|f(x_1) - f(x_2)| < (x_1 - x_2)^2$, for all $x_1, x_2 \in \mathbb{R}$. Find the equation of tangent to the curve $y = f(x)$ at the point (1, 2).
8. Find the equation of the plane containing the line $2x - y + z - 3 = 0$, $3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point (2, 1, -1)
9. Find the area bounded by the curve $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$.

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10. If one of the vertices of the square circumscribing the circle $|z - 1| = \sqrt{2}$ is $2 + \sqrt{3}i$. Find the other vertices of the square.
11. If $f(x - y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ and $g(x - y) = g(x) \cdot g(y) + f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$ and right hand derivative at $x = 0$ exists for $f(x)$, then find derivative of $g(x)$ at $x = 0$.
12. If length of tangent at any point on the curve $y = f(x)$ intercepted between the point and the x-axis is of length 1. Find the equation of the curve.
13. Find the equation of the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$. Also find the length of the intercept AB of the tangent between the coordinate axes.
14. Tangents are drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 9$. Find the locus of mid-point of the chord of contact.
15. If $P(x)$ be a polynomial of degree 3 satisfying $P(-1) = 10$, $P(1) = -6$ and $P(x)$ has maxima at $x = -1$ and $P'(x)$ has minima at $x = 1$. Find the distance between the local maxima and local minima of the curve.
16. If the incident ray on a surface is along the unit vector $\hat{v}$, the reflected ray is along the unit vector $\hat{w}$ and the normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.
17. If $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2 + 3a \\ 3b^2 + 3b \\ 3c^2 + 3c \end{bmatrix}$, $f(x)$ is a quadratic function and its maximum value occurs at a point V. A is a point of intersection of $y = f(x)$ with x-axis and point B is such that chord AB subtends a right angle at V. Find the area enclosed by $y = f(x)$ and chord AB.
18. $f(x)$ is differentiable function and $g(x)$ is a double differentiable function such that $|f(x)| \leq 1$ and $f'(x) = g(x)$. If $f^2(0) + g^2(0) = 9$ then prove that there exists some $c \in (-3, 3)$ such that $g(c) \cdot g''(c) < 0$.

ANSWER KEY TO IIT-JEE 2005 PAPER

MAIN EXAMINATION

1. $\sqrt{5}$

3. 7

5. $\left[-\frac{\pi}{2}, -\frac{\pi}{10}\right] \cup \left[\frac{3\pi}{10}, \frac{\pi}{2}\right]$

7. $y = 2$

9. $\frac{1}{3}$ square units

11. 0

13. $AB = \frac{14}{\sqrt{3}}$

15. $4\sqrt{65}$

17. $\frac{125}{3}$ sq.units

2. $\frac{24}{5} \left[e \cos\left(\frac{1}{2}\right) + \frac{1}{2} e \sin\left(\frac{1}{2}\right) - 1 \right]$

4. $1/7$

6. $(y-1)^2 = 4x^2, y \neq 1$

8. $2x - y + z - 3 = 0, 62x + 29y + 19z - 105 = 0$

10. $-i\sqrt{3}, 1 - \sqrt{3} + i, 1 + \sqrt{3} - i$

12. $\log \left| \frac{1 - \sqrt{1-y^2}}{y} \right| + \sqrt{1-y^2} = \pm x + c$

14. $\frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2 + y^2}{9} \right)^2$

16. $\hat{w} = \hat{v} - 2(\hat{a} \cdot \hat{v})\hat{a}$